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The relationship between peak height velocity and physical performance in youth soccer players

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Abstract

Longitudinal changes in height, weight and physical performance were studied in 33 Flemish male youth soccer players from the Ghent Youth Soccer Project. The players' ages at the start of the study ranged from 10.4 to 13.7 years, with a mean age of 12.2 ± 0.7 years. Longitudinal changes were studied over a 5 year period. Peak height velocity and peak weight velocity were determined using non-smoothed polynomials. The estimations of peak height velocity, peak weight velocity and age at peak height velocity were $9.7 \pm 1.5\,\mathrm{cm \cdot year^{-1}}$, $8.4 \pm 3.0\,\mathrm{kg \cdot year^{-1}}$ and 13.8 ± 0.8 years, respectively. Peak weight velocity occurred, on average, at the same age as peak height velocity. Balance, speed of limb movement, trunk strength, upper-body muscular endurance, explosive strength, running speed and agility, cardiorespiratory endurance and anaerobic capacity showed peak development at peak height velocity. A plateau in the velocity curves was observed after peak height velocity for upper-body muscular endurance, explosive strength and running speed. Flexibility exhibited peak development during the tear after peak height velocity. Trainers and coaches should be aware of the individual characteristics of the adolescent growth spurt and the training load should also be individualized at this time.

Keywords: *Maturation, youth sports, peak height velocity, growth rate*

Introduction

Physical performance is related to biological maturation during male adolescence. The relationship is more pronounced when boys of contrasting maturity status (i.e. early vs. late maturers) are compared. Boys who are advanced in biological maturity are generally better performers than their later maturing peers (Beunen, Ostyn, Simons, Renson, & Van Gerven, 1980a; Beunen *et al.*, 1988; Malina, Bouchard, & Bar-Or, 2004).

These inter-individual differences in performance are generally transient as late maturers, on average, catch up in many aspects of performance in young adulthood (Lefevre, Beunen, Steens, Claessens, & Renson, 1990). Nevertheless, several studies have reported that increased selection opportunities in soccer tend to favour older and physically taller boys (Brewer, Balsom, & Davis, 1995; Simmons & Paull, 2001), so that proportionally fewer later maturing boys are represented on soccer teams after 13 years of age (Malina, 2003). Furthermore, players born early in the competition year tend to dominate national soccer leagues (Dudink, 1994; Musch & Hay, 1999).

The adolescent growth spurt varies considerably in timing, tempo and duration among individuals. Allowing for this variation, peak height velocity rather than chronological age has been used to characterize changes in size, body composition and performance relative to the adolescent spurt in height (Beunen & Malina, 1988; Beunen *et al.*, 1988; Malina *et al.*, 2004). Cross-sectional data are reasonably consistent in showing that early maturing boys tend to be more successful in soccer in mid- and late adolescence. From about 14 years of age, boys advanced in maturity status (sexual and skeletal maturation) are better represented on youth soccer teams (Cacciari *et al.*, 1990; Malina, 2003; Malina *et al.*, 2000; Peña Reyes, Cardenas-Barahona, & Malina, 1994). Corresponding data for the age at peak height velocity among adolescent soccer players

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are limited. Studies of Welsh (Bell, 1993) and Danish (Froberg, Anderson, & Lammert, 1991) youth soccer players indicated identical ages at peak height velocity (i.e. 14.2 ± 0.9 years). This value is well within the range of estimated ages at peak height velocity for samples of European boys (i.e. 13.8–14.2 years; Malina *et al.*, 2004) and suggests "average" maturity status.

Data for the general population of adolescent males suggest that strength and power attain maximal growth after peak height velocity, running speed attains maximal growth before peak height velocity, and maximal aerobic power reaches maximal growth coincident with peak height velocity (Beunen & Malina, 1988; Malina *et al.*, 2004). Corresponding information on longitudinal changes in strength and motor performance relative to peak height velocity in adolescent athletes and in particular soccer players is not available. The aims of this study were to examine the adolescent growth spurt in height and the development of physical performance relative to the time of maximal growth in height in male soccer players, and to make comparisons with data from the general population of male adolescents. We hypothesize that age at peak height velocity and timing of development of physical performance relative to it occur earlier in youth soccer players compared with a general population of adolescent males.

Methods

Participants

The data are from the Ghent Youth Soccer Project (GYSP), a 5 year mixed-longitudinal study on growth and performance of soccer players. The GYSP was carried out between 1996 and 2000 during the months of February, March, April and May of each year. The Ethics Committee of the Ghent University Hospital approved the study, and informed consent was obtained from the players and their parents before the study geban. The mixed-longitudinal sample included 232 youth soccer players of different competitive standards (elite, sub-elite, non-elite) from 10 soccer clubs in Flanders. Clubs were randomly selected from the provinces of Antwerp, West-Flanders and East-Flanders. After approval from the clubs' executive committee, youth teams representing the appropriate age range were asked to participate in the study. Elite players participated in 6 h of combined competitive play and soccer training per week on average (consisting of 4–5 sessions). Sub-elite and non-elite players participated in 4 h (3 sessions a week) and 3 h (2 sessions a week) per week, respectively. At the start of the GYSP, for each club one youth team with players aged 11–13 years was asked to participate. Of the total sample, 51 players were measured annually over five consecutive years and 25 players were measured annually over 4 years, resulting in 76 potential participants for the present analysis. The players' ages at the start of the study ranged from 10.4 to 13.7 years, with a mean of 12.2 ± 0.7 years.

Procedures

Height was measured with a fixed stadiometer to the nearest 0.1 cm and weight was measured with a Seca beam balance to the nearest 0.1 kg. The Eurofit test of physical fitness (Council of Europe, 1988) and several soccer-specific performance tests were administered (Table I). The Eurofit items included the flamingo balance (FBA), bent arm hang (BAH), standing long jump (SLJ), sit-ups (SUP), 10 × 5 m shuttle run (SHR), plate tapping (PLT), sit-and-reach (SAR) and endurance shuttle run (ESHR). The hand grip test was not used in the present analysis. The soccer-specific items included two measures of running speed – the 30 m dash (DASH, best of three attempts, flying start and hand-held chronometry) and the 5 × 10 m shuttle sprint (SSPRINT, best of two attempts) – 'explosive strength' as measured by the vertical jump (VTJ, best of three attempts) and anaerobic capacity as measured by a shuttle run (STEMPO), in which the participants ran 300 m as fast as possible between markers following the given distances: 2 × 10 m, 2 × 20 m, 2 × 30 m, 2 × 40 m, 2 × 50 m. Anthropometric and fitness characteristics were measured by two experienced test leaders (M.J. and B.V.R.) throughout the longitudinal study to ensure high-quality data collection. The Eurofit test battery and the vertical jump were performed indoors following the normal guidelines and test descriptions (Claessens, Vanden Eynde, Renson, & Van Gerven, 1990; Council of Europe, 1988). The soccer-specific

Table I. Physical performance factors and their corresponding tests.

Factor	Test
Total body balance	Flamingo balance (FLB)
Speed of limb movement	Plate tapping (PLT)
Flexibility	Sit and reach (SAR)
Trunk strength	Sit-ups (SUP)
Upper-body muscular endurance	Bent arm hang (BAH)
Explosive strength	Standing long jump (SLJ) Vertical jump (VTJ)
Running speed/agility	Shuttle run: 10 × 5 m (SHR) Shuttle spurt: 5 × 10 m (SSPRINT)
Running speed	30 m dash (DASH)
Cardiorespiratory endurance	Endurance shuttle run (ESHR)
Anaerobic capacity	Shuttle tempo (STEMPO)

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tests were performed outside on the soccer field (Verheijen, 1997). The players wore their usual soccer clothing and shoes. One complete test day per soccer team was scheduled.

Individual data for height and each performance test were fitted with the modified non-smoothed polynomial method used previously in studies of adolescent growth and performance (Beunen *et al.*, 1988; Yague & De La Fuente, 1998). This method was used because of the similarity between the present data and those of the studies of Belgian (Beunen *et al.*, 1988) and Spanish (Yague & De La Fuente, 1998) boys.

Although measurements were taken once annually (M1 to M5), the modified non-smoothed polynomial method (Figure 1) permits estimation of velocities for half-year intervals (V1 to V7) at the considered time points (T1 to T7). First, four annual increments (I1, I2, I3 and I4) were calculated and corrected for deviations from the one-year intervals. Then, velocity values at 6-month intervals were estimated using polynomials of second (V2, V4, V6) and third degree (V1, V3, V5, V7).

Velocity curves for height were checked graphically to determine the location of the spurt, and peak height velocity was defined as the highest velocity recorded. Boys whose maximal velocity was located at V1 or V7 (or V5 for those who were examined four times) were excluded because it is likely that the real maximal velocity was located before V1 or after V7 (or V5).

Individual velocity curves for height and the performance variables were aligned on peak height velocity. Mean velocity curves were calculated and defined in terms of months before or after peak height velocity. Based on the calculated velocity for each half year, age at peak velocity for height and each performance variable could be estimated. A more detailed description of the method is given in Beunen *et al.* (1988).

Since all participants were tested in at least four consecutive years, a learning effect due to test participation is a possible confounding factor in estimating growth velocities (Beunen, Simons, Ostyn, Renson, & Van Gerven, 1980b). This effect was checked by comparing the mean values for all variables under study from this sample with the mean values from the GYSP participants of the same chronological age who participated only once or twice (van Mechelen & Mellenbergh, 1997). Mean values did not differ significantly between the two groups.

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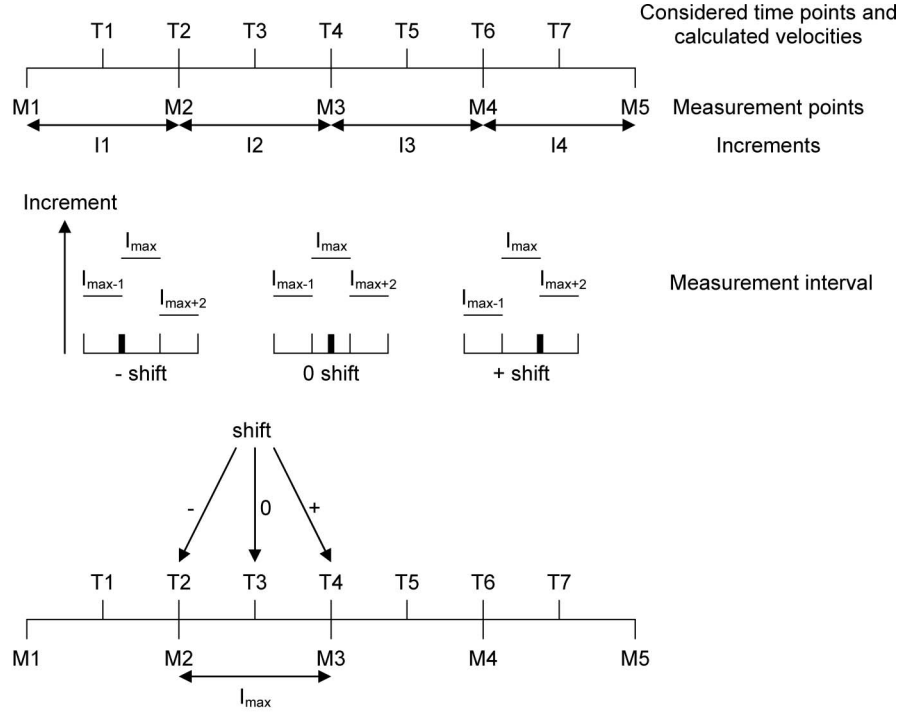

Figure 1. Diagram of the adapted non-smoothed polynomials method for the calculation of growth velocities, and location of the age at spurt starting from the maximal increment, I_{max}.

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Skeletal maturation was assessed in the GYSP by a paediatrician from the University Hospital. X-rays of the left hand and wrist were taken on the same test day. Skeletal age was estimated according to the TW2 method (Tanner *et al.*, 1983) as a reference for the peak height velocity data.

Results

The modified non-smoothed polynomial method was successfully fitted to height and weight for 33 and 21 soccer players, respectively (mean age and skeletal age at start of the study were 12.1 ± 0.7 and 12.4 ± 1.3 years, respectively). Mean ages at peak height velocity and peak weight velocity were identical (i.e. 13.8 ± 0.8 years). Peak velocities for height and weight were $9.7 \pm 1.5\,\mathrm{cm \cdot year^{-1}}$ and $8.4 \pm 3.0\,\mathrm{kg \cdot year^{-1}}$, respectively (Table II). Height velocity estimates for all 21 players with estimated velocity for weight are included in the sample of 33 players.

The individual height records of 43 boys could not be modelled successfully with the modified non-smoothed polynomial. Of these players with four or five longitudinal observations, age at peak height velocity was attained by 25 boys before the start of the study (mean age and skeletal age at start of the study were 12.6 ± 0.5 and 13.5 ± 1.2 years respectively), whereas peak height velocity had not yet been attained by 18 boys at the end of the study period (mean age and skeletal age at start of the study were 11.6 ± 0.8 and 11.1 ± 1.1 years respectively). The mean constant velocity curve for the heights and weights of the 33 and 21 boys whose records were successfully modelled by the modified non-smoothed polynomial are shown in Figure 2.

Mean constant velocity curves for the performance tests are illustrated in Figures 3 and 4, while mean velocities are summarized in Table III. Maximal velocities for balance (FBA, 2.5 attempts $\cdot$ year^{-1}; Figure 3A), speed of limb movement (PLT, $0.8\,\mathrm{s \cdot year^{-1}}$; Figure 3B) and trunk strength (SUP, 2.7 sit-ups $\cdot$ year^{-1}; Figure 3D) were achieved, on average, simultaneously with peak height velocity; after the peak, velocities of development in each task declined linearly. Maximal velocity for trunk flexibility (SAR, $2.7\,\mathrm{cm \cdot year^{-1}}$; Figure 3C) was attained one year after peak height velocity. The pattern of gain in trunk flexibility before peak height velocity indicates a declining trend which reached its nadir at peak height velocity ($-0.1\,\mathrm{cm \cdot year^{-1}}$).

Velocity of growth in upper-body muscular endurance (BAH) increased gradually from 18 to 12 months before peak height velocity ($-2.9\,\mathrm{s \cdot year^{-1}}$) and then increased more sharply to a peak coincident with peak height velocity ($7.6\,\mathrm{s \cdot year^{-1}}$; Figure 4A). The velocity of development in upper-body muscular endurance gradually declined towards 12 months after peak height velocity ($5.2\,\mathrm{s \cdot year^{-1}}$).

The two "explosive strength items", standing long jump (SLJ) and vertical jump (VTJ), showed different patterns of growth velocity during adolescence. Maximal increase in explosive strength (SLJ, $10.5\,\mathrm{cm \cdot year^{-1}}$) was reached, on average, almost 18 months before peak height velocity. Subsequently, estimated velocities for the standing long jump declined to 12 months before peak height velocity ($6.3\,\mathrm{cm \cdot year^{-1}}$) and then gradually increased during the interval between 12 months before and 12 months after peak height velocity, so that the estimated velocity at this time is quite similar to the earlier peak ($10.1\,\mathrm{cm \cdot year^{-1}}$; Figure 4B). In contrast, explosive strength measured as the vertical jump showed a similar estimated velocity curve to that for height with a velocity of $1.5\,\mathrm{cm \cdot year^{-1}}$ 12 months before peak height velocity and reaching a peak coincident with peak height velocity (VTJ, $5.1\,\mathrm{cm \cdot year^{-1}}$; Figure 4B). Subsequently, the estimated velocity of the vertical jump declined to $3.3\,\mathrm{cm \cdot year^{-1}}$ one year after peak height velocity.

The estimated velocity curves for the three measures of running speed showed a similar pattern across the growth spurt in height (Figure 4C). The curve for the 10×5 m shuttle run (SHR) is symmetric. It increased from 12 months before peak height velocity ($0.4\,\mathrm{s \cdot year^{-1}}$), reached a peak coin-

Table II. Mean growth velocities for height and weight when individual data of soccer players are aligned on their respective individual peak height velocity (PHV).

		Months from PHV								
Variables		−24	−18	−12	−6	0	6	12	18	24
Height (cm · year^{-1})	mean	5.7	5.7	6.7	8.2	9.7	7.6	5.6	4.5	3.5
	s	1.1	0.9	1.7	1.3	1.5	1.2	2.4	1.9	1.6
	n	9	9	33	33	33	33	33	17	17
Weight (kg · year^{-1})	mean	2.7	3.2	5.0	6.6	8.4	7.1	6.4	5.7	4.1
	s	0.8	1.6	2.1	2.0	3.0	1.9	3.1	2.3	3.1
	n	4	4	21	21	21	21	21	14	14

Note: Mean age at PHV = 13.8 years.

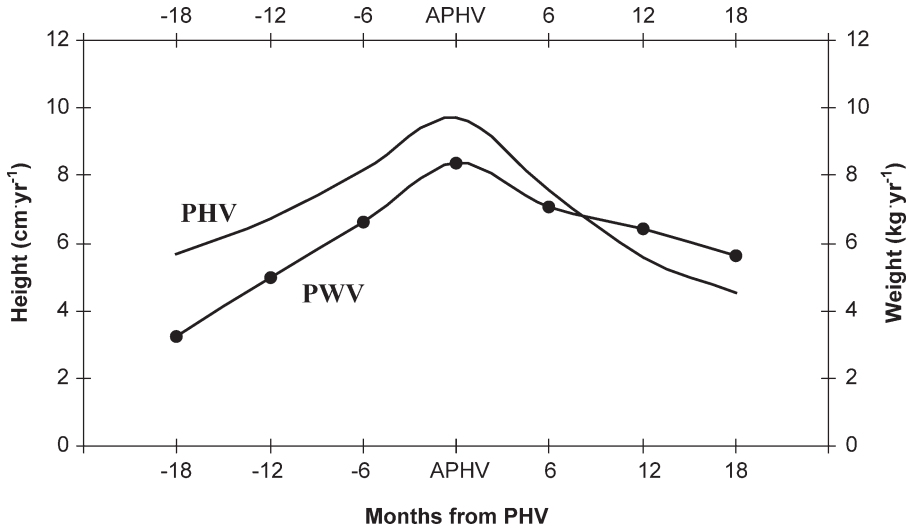

Figure 2. Mean constant velocity curve for height (PHV) and weight (PWV) in soccer players (PHV = 9.7 cm · year^{-1}, PWV = 8.4 kg · year^{-1}).

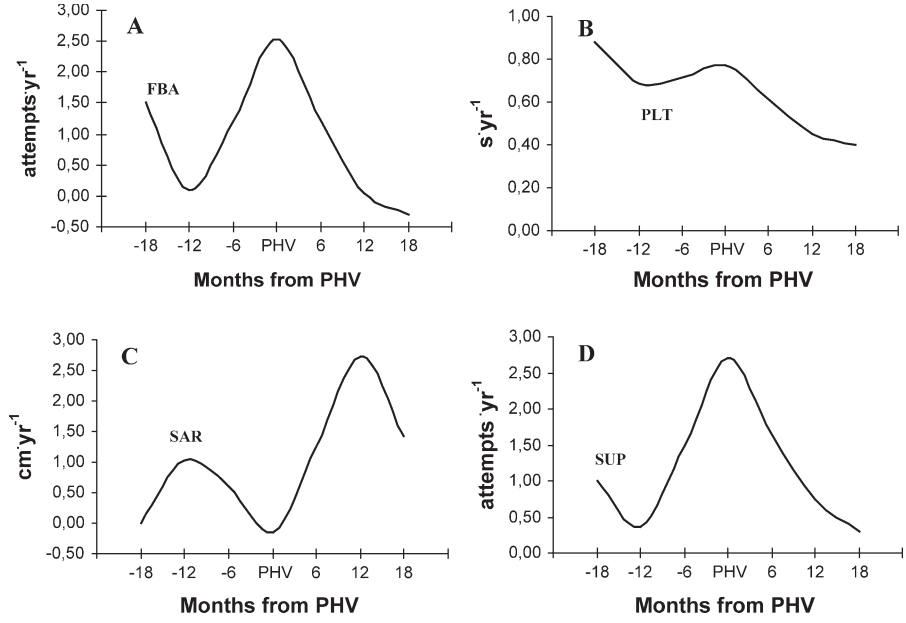

Figure 3. Mean constant velocity curve for (A) flamingo balance (FBA, attempts · year^{-1}), (B) plate tapping (PLT, s · year^{-1}), (C) sit and reach (SAR, cm · year^{-1}) and (D) sit-ups (SUP, attempts · year^{-1}).

cident with peak height velocity (1.6 s · year^{-1}) and then decreased to 12 months after peak height velocity (0.1 s · year^{-1}). The curve for the 5 × 10 m shuttle run (SSPRINT), though not as intense, was similar, reaching a peak gain coincident with peak height velocity (0.9 s · year^{-1}). The estimated velocity curve for the 30 m dash (DASH) showed negative values for the interval before peak height velocity (− 0.6 s · year^{-1} 12 months before peak height velocity), but subsequently moved towards positive values and reached a peak at peak height velocity (0.4 s · year^{-1}). Subsequently, the velocity curve for the 30 m dash showed a plateau for 12–18 months after peak height velocity.

Estimated velocities for cardiorespiratory endurance measured as an endurance shuttle run (ESHR) showed a symmetric pattern relative to peak height velocity. Velocities increased, on average, from 12 months before peak height velocity (0.1 min · year^{-1}), reached a maximum coincident with peak height velocity (1.5 min · year^{-1}), and then declined to 0.0 min · year^{-1} (ESHR, Figure 4D).

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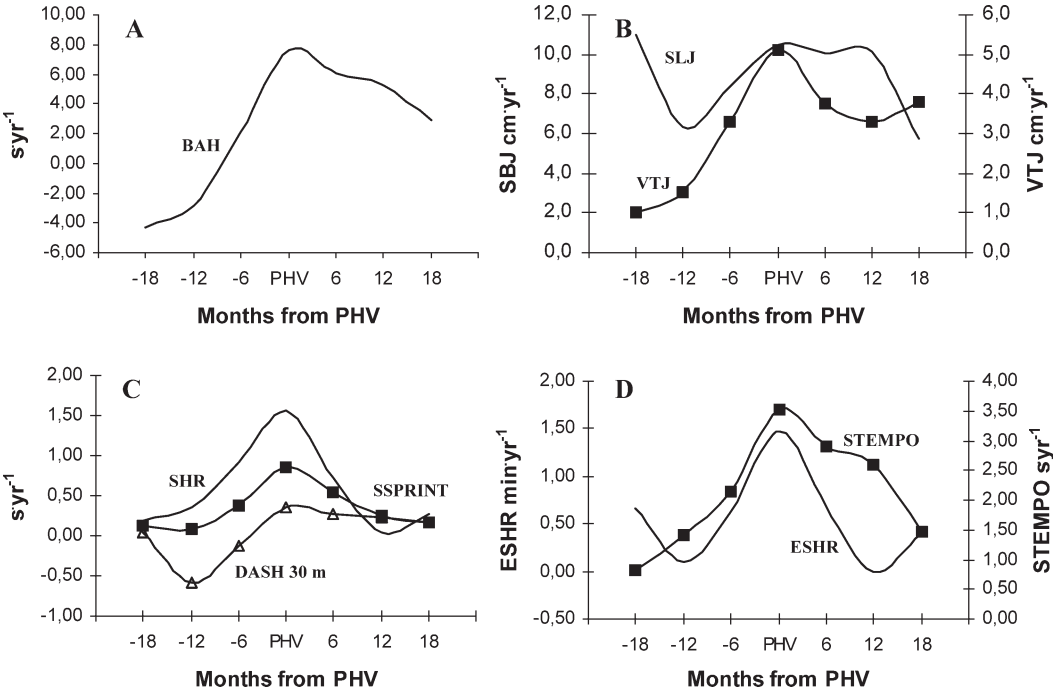

Figure 4. Mean constant velocity curve for (a) bent arm hang (BAH, $s \cdot year^{-1}$), (B) standing long jump (SLJ) and vertical jump (VTJ, $cm \cdot year^{-1}$), (C) shuttle run (SHR), shuttle sprint (SSPRINT) and 30 m dash (DASH, $s \cdot year^{-1}$), and (D) endurance shuttle run (ESHR, $min \cdot year^{-1}$) and shuttle tempo (STEMPO, $s \cdot year^{-1}$).

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Corresponding velocities for anaerobic capacity measured as a paced shuttle run increased from 18 months before to a peak at peak height velocity; however, the estimated velocity declined only slightly after peak height velocity, suggesting a plateau with velocities of $3.5\,s \cdot year^{-1}$ and $2.9\,s \cdot year^{-1}$ at and 6 months after peak height velocity respectively (STEMPO, Figure 4D). Estimated velocities for anaerobic capacity measured as a paced shuttle run then declined.

Discussion

Longitudinal studies of the growth, maturation and performance of youths participating in specific sports are limited. The Ghent Youth Soccer Project is a mixed-longitudinal study of a combined sample of elite, sub-elite and non-elite players aged 10–16 years. The estimated mean age at peak height velocity for 33 players was 13.8 ± 0.8 years. It is somewhat earlier than estimates for samples of Welsh (Bell, 1993) and Danish (Froberg *et al.*, 1991) youth soccer players (i.e. 14.2 ± 0.9 years), but is well within the range of estimated ages at peak height velocity for samples of European boys (13.8–14.2 years; Malina *et al.*, 2004). However, the estimated age at peak height velocity is probably earlier in the sample of players in the Ghent Youth Soccer Project because individual data plots for 25 boys suggested that they had already attained peak height velocity before the start of the study. On the other hand, an examination of the data for another 18 boys suggested that they had not yet reached peak height velocity at the end of the study. A comparison of the boys' skeletal ages confirmed this suggestion. At the start of the study, the skeletal age of the three groups differed significantly ($P < 0.05$) from each other: 12.4 ± 1.3 years for the sample as a whole, 13.5 ± 1.2 years for the 25 boys that attained peak height velocity before the start of the study, and 11.1 ± 1.1 years for the 18 boys that had not reached peak height velocity at the end of the study. These data are consistent with other estimates of maturity status in elite soccer players – that is, advanced skeletal and sexual maturity status compared with non-athletic boys (Malina, 2003; Malina *et al.*, 2000). The estimated average peak height velocity of the soccer players ($9.7\,cm \cdot year^{-1}$) is also well within the range (8.2 to $10.3\,cm \cdot year^{-1}$) for European boys (Malina *et al.*, 2004). Allowing for the limited number of observations per player and the use of modified non-smoothed polynomials (Beunen *et al.*, 1988; Yague & De La Fuente, 1998), the estimates are reasonably consistent with other data for the age at peak height velocity and for peak height velocity.

However, application of the modified non-smoothed polynomial method to the performance

Table III. Mean growth velocities in different physical fitness tests when individual data are aligned on peak height velocity (PHV).

		Months from PHV				
Motor tests		−12	−6	0	6	12
FBA (attempts · year^{-1})	mean	0.1	1.2	**2.5**	1.2	0.1
	n	12	12	12	12	10
PLT (s · year^{-1})	mean	0.7	0.7	**0.8**	0.6	0.5
	n	9	9	9	9	8
SAR (cm · year^{-1})	mean	1.0	0.6	−0.1	1.3	**2.7**
	n	9	9	8	8	8
SLJ (cm · year^{-1})	mean	6.3	8.4	**10.5**	10.1	10.1
	n	10	10	10	10	10
VTJ (cm · year^{-1})	mean	1.5	3.3	**5.1**	3.8	3.3
	n	12	12	12	12	12
SUP (sit-ups · year^{-1})	mean	0.4	1.5	**2.7**	1.6	0.8
	n	11	11	11	11	10
BAH (s · year^{-1})	mean	−2.9	2.2	**7.6**	6.1	5.2
	n	17	17	16	16	16
SHR (s · year^{-1})	mean	0.4	0.9	**1.6**	0.7	0.1
	n	14	14	14	14	12
SSPRINT (s · year^{-1})	mean	0.1	0.4	**0.9**	0.5	0.2
	n	12	12	10	10	9
DASH (s · year^{-1})	mean	−0.6	−0.1	**0.4**	0.3	0.2
	n	8	8	8	8	7
ESHR (min · year^{-1})	mean	0.1	0.6	**1.5**	0.7	0.0
	n	12	12	11	11	9
STEMPO (s · year^{-1})	mean	1.4	2.1	**3.5**	2.9	2.6
	n	15	15	14	15	14

Note: Number of participants (*n*) can vary between tests at successive half-year intervals before and after age at PHV. Boys whose maximal velocity points were located at V1 or V7 (or V5 for those who were examined four times) were excluded because it is likely that the real maximal velocity was located before V1 or after V7 (or V5).
Abbreviations: FBA = flamingo balance; PLT = plate tapping; SAR = sit and reach; SLJ = standing long jump; VTJ = vertical jump; SUP = sit-ups; BAH = bent arm hang; SHR = 10 × 5 m shuttle run; SSPRINT = 5 × 10 m shuttle sprint; DASH = 30 m dash; ESHR = endurance shuttle run; STEMPO = anaerobic capacity as measured by a shuttle run.

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items results in more variable estimates of velocity curves. The estimates are also influenced by the drop-out of some players and missing data for some items. Allowing for these caveats, estimated velocities for the performance items aligned on peak height velocity are summarized in Table IV. It appears that most of the performance items reach a mean peak velocity that is coincident with the timing of peak height velocity.

Data for the general population of adolescent males suggested that maximal gains in muscular strength and power occur, on average, after peak height velocity and closer to peak weight velocity (Beunen & Malina, 1988; Malina *et al.*, 2004). This is probably related to the adolescent spurt in muscle mass that occurs shortly after peak height velocity (Malina *et al.*, 2004). The data for youth soccer players indicated peak gains in bent arm hang and vertical jump performance coincident with peak height velocity; however, estimated velocities for these items (and also for standing long jump) remained positive for some time after peak height velocity. The trends for muscular strength and power reflect continued growth and perhaps an influence of systematic sport training.

Trunk strength of the soccer players reached a maximal velocity coincident with peak height velocity (2.7 sit-ups · year^{-1}), which contrasts with observations for a longitudinal sample of Belgian boys (Beunen *et al.*, 1988). Differences in methodology may explain this variation; the study of Belgian boys used leg lifts completed in 20 s as the indicator of trunk strength. Ellis, Carron and Bailey (1975) and Yague and De La Fuente (1998) also observed a growth spurt in trunk strength, but it occurred, on average, after the spurt in height. The estimated maximal gain in trunk strength in the soccer players was smaller than the estimated peak gain in Spanish boys (4.6 sit-ups · year^{-1}) using the same modified non-smoothed polynomial method (Yague & De La Fuente, 1998). However, trunk strength of the present soccer players continued to improve during the observation period in contrast to the clear drop in velocity in the Spanish sample. The continuous

Table IV. Timing and peak velocity for general and specific motor tests in soccer players.

Motor tests	Timing	Peak velocity (unit · year^{-1})
FBA	At moment of PHV	2.5 attempts
PLT	At moment of PHV	0.8 s
SAR	12 months after PHV	2.7 cm
SUP	At moment of PHV	2.7 sit-ups
BAH	At moment of PHV with plateau	7.6 s
SLJ	At moment of PHV with plateau	10.5 cm
VTJ	At moment of PHV	5.1 cm
SHR	At moment of PHV	1.6 s
SSPRINT	At moment of PHV	0.9 s
DASH	At moment of PHV with plateau	0.4 s
ESHR	At moment of PHV	1.5 min
STEMPO	At moment of PHV	3.5 s

Abbreviations: PHV = peak height velocity; FBA = flamingo balance; PLT = plate tapping; SAR = sit and reach; SLJ = standing long jump; VTJ = vertical jump; SUP = sit-ups; BAH = bent arm hang; SHR = 10 × 5 m shuttle run; SSPRINT = 5 × 10 m shuttle sprint; DASH = 30 m dash; ESHR = endurance shuttle run; STEMPO = anaerobic capacity as measured by a shuttle run.

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improvement in trunk strength might be related to the training status of the participants. They undertook at least 3 h per week of soccer training and a competitive game.

Estimated peak velocities for speed-related tasks were attained somewhat before (plate tapping) or coincident with (10 × 5 m shuttle run, 5 × 10 m shuttle sprint) peak height velocity in the soccer players. These trends contrast with those for the shuttle run in Belgian and Spanish boys, among whom estimated peak velocities occurred 18 months (Beunen *et al.*, 1988) and 8 months (Yague & De La Fuente, 1998) before peak height velocity, respectively. The data for Spanish boys, however, indicated a second period of rapid development after peak height velocity (Yague & De La Fuente, 1998), but the significance of this observation is not clear. Performance of the soccer players in the 30 m dash showed an inverse relationship with height growth in the year before peak height velocity, but estimated velocities became and remained positive (improving performance) at peak height velocity. Most of the players who showed a decline in 30 m dash performance before peak height velocity were generally moderate to good performers at the beginning of the peak height velocity interval, compared with those who showed consistent positive velocities. These findings are in accordance with the results of Beunen *et al.* (1988). This phenomenon of a temporary decline in performance or a disruption of motor coordination is called the "adolescent awkwardness". Because of its temporary character, the awkwardness tends to depend more on individual patterns and changes in growth and performance (Beunen *et al.*, 1988; Butterfield, Lehnhard, Lee, & Cloadarci, 2004).

The estimated velocity curve for lower-back flexibility in the soccer players suggests a peak one year after peak height velocity, which contrasts with trends observed in earlier studies of Belgian (Beunen *et al.*, 1988) and Canadian (Ellis *et al.*, 1975) boys. However, data for Spanish boys (Yague & De La Fuente, 1998) indicated a similar trend to that observed in the soccer players and an inverse relationship between the development of flexibility and growth in height for a short period around peak height velocity. The temporarily limited development of lower-back flexibility (with a negative velocity at peak height velocity) may, in part, be related to the differential timing of adolescent spurts in leg and trunk length. The legs reach peak velocity before peak height velocity, whereas the trunk reaches peak velocity after peak height velocity (Malina *et al.*, 2004). It is possible that the later growth in lower-back flexibility is associated with late adolescent growth in the trunk (and also arm length).

The estimated velocity curves for the cardiorespiratory endurance tests, which indicate peak gains at peak height velocity, differ from those observed for Spanish boys (Yague & De La Fuente, 1998), among whom maximal velocity occurred 8 months after peak height velocity followed by a sharp decline in velocities. Corresponding data for peak oxygen uptake indicated maximal gains coincident with peak height velocity and continuous improvement during adolescence (Mirwald & Bailey, 1986). This would suggest that better performances in the endurance shuttle run are related to improved maximal oxygen uptake and not only to improved running economy as suggested by Yague and De La Fuente (1998). Data from this study of youth soccer players also showed no differences in running economy between early and late maturing boys, even after using allometric scaling for body mass. Nevertheless, running style appears to be an important

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determinant of running economy (Segers, De Clercq, Philippaerts, & Janssens, 2002).

The anaerobic capacity of the soccer players also showed a spurt at the time of peak height velocity. However, the estimated velocity curve indicated continued to improve after peak height velocity. This is consistent with general observations that anaerobic performance probably improves into late adolescence (Bar-Or, 1983; Malina *et al.*, 2004).

A relevant question is whether there is a sensitive period for training the different physical fitness components. Is it possible to enhance physical performance by training during the growth spurts? Should soccer-specific training be implemented at a particular maturational stage – that is, before, at or after peak height velocity? These questions need further longitudinal study in combination with experimental programmes to examine possible (long-term) effects of different training regimes. Experimental studies indicate that strength and aerobic capacity in adolescent boys are less trainable in absolute terms, but are as trainable and perhaps more trainable in relative terms compared with young adults, if the training stimuli are adequate and appropriate (Blimkie *et al.*, 1996; Pate & Ward, 1996; Pfeiffer & Francis, 1986; Sale, 1989). The trainability of other components of physical performance during adolescence is a complex matter. Anaerobic capacity during high-intensity exercise seems to be less developed in pre-adolescents than adults (Bar-Or, 1983; Paterson, Cunningham, & Bumstead, 1986). However, data pertaining to the effects on capacity and endurance are inadequate to draw conclusions about exercise guidelines for children and pre-adolescents. Therefore, it is reasonable to be cautious in developing and implementing training regimes for the enhancement of functional characteristics and sport-specific skills of children and adolescents.

An important implication for youth soccer is that individual growth velocities should be taken into account. As soccer is a team sport, players from the same team can differ considerably in physique due to their individual pace in development. As a consequence, late-maturing boys appear to be systematically excluded from many team sports (Malina, 2003; Simmons & Paull, 2001). Thus, in the context of talent identification and development, trainers and coaches should be aware of the characteristics of the adolescent growth spurt and recognize that changes in growth and performance at this time are highly individualized. In the context of planning long-term talent development, the best performers in adulthood were not necessarily early maturing boys. Those who performed well for their maturity level during adolescence had a good chance of still performing above average at the age of 30 (Lefevre *et al.*, 1990).

In conclusion, estimated velocities for most performance tests reached a peak around the time of maximal growth in height. In many tasks, however, performances continued to improve after peak height velocity, probably reflecting differential timing of growth in muscle mass and perhaps the influence of systematic soccer-specific training. Differences in velocities between soccer players in this study and the general population of non-athletic adolescent boys tend to be small. It is apparent that athletes and non-athletes experience adolescent growth spurts in body size and performance capabilities, but variation in the timing and tempo of maximal growth during the adolescent spurt is real. The data do not permit the partitioning of expected growth-related changes from those that might be associated with soccer-specific training. Further longitudinal research is needed to address this issue.

There is a need for further prospective studies that follow sufficiently large samples of young athletes from late childhood through puberty. Moreover, information about nutritional status, sport training and injury history, activity level, hormonal secretions and perhaps genetic markers to identify genotype–environment interactions should be included in such studies.

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